Climate Change and Human Displacement Forecasting

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Abstract

Climate change is known to drive human displacement. The question that drove inspiration for this project was to see if machine learning could help forecast future climate related migration flows. We have three main datasets that give us refugee displacement records, disaster impacts, and climate observations. Once aggregated and preprocessed, we then used the dataset to train 3 models: a single decision tree, Random Forest, and XGBoost. We also used a time based split, so we trained on 2000-2015 and tested on 2016-2019. This was to ensure realistic evaluation for our models. Random Forest and XGBoost achieved the best performance. We had to evaluate in log space because of the heavy-tailed distribution of displacement events. What we found through this project are patterns similar to prior climate-migration research, but we also added a predictive dimension. We developed an interactive tool that allows the public to explore 2030 displacement under different climate scenarios.

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1 Introduction (1 page)

Climate change is rapidly coming to the forefront of global issues that we are facing today. It has led to more frequent extreme weather events, which in turn causes more people to be displaced.

This project explores the key question of: *Can we use machine learning as a tool to predict how climate change and natural disaster would affect future migration patterns?* We did not want to just focus on what has happened in the past and build a model that analyzes it. We wanted to build models that allowed us to see what the future has in store with respect to climate migration. We realized that building a model like this could help governments and other global institutions prepare for climate related migration before they even occur. If we can identify potential hotspots of future displacement, then they can better allocate their resources and start devising strategies in advance.

To do this, we collected and merged multiple global datasets: refugee displacement records (UNHCR), natural disaster impacts (IDMC), and climate variables (CRU/NASA). With the data, we trained several machine learning models (decision trees and ensemble methods) to predict country displacement over time.

1.1 Research Question(s)

Our main questions are:

- Can machine learning models predict how many people will be displaced in a given country-year based on climate and disaster data?
- Which variables are the strongest predictors of future displacement?
- How can these predictions be used to support climate adaptation, governmental planning, and policy development?

1.2 Target Audience

This project is intended for a pretty broad, but also informed audience. Anywhere from climate policy makers to high school or university students interested in applied machine learning.

We aimed to present our findings in an intuitive and easily accessible way.

- **Policy makers and governmental planners** that want models to better make informed decisions on where to allocate funds with respect to potential climate refugees.
- **Climate scientists and researchers** who are exploring connections between environmental change and social impacts.
- **Machine learning hobbyists and students** interested in applying predictive modeling to real world global challenges.

2 Related Work (0.5-1 page)

Impacts on Vulnerable Populations. Benevolenza and DeRigne (2019) conducted a global review *"The impact of climate change and natural disasters on vulnerable populations"* [Benevolenza and DeRigne, 2019]. In short, what the authors focused on is whether or not low income families, the elder, and other minority communities suffer more than others due to climate related events. They argue that these groups face higher probability of physical and mental health issues after disasters, so it would make sense that there would be an uneven toll of climate change. What we thought about this paper is that the topic is very interesting and similar to ours, but it is more descriptive than what we wanted to do. It does not use any models or attempts to predict future displacement. So we wanted to build on their understanding of vulnerability, but extend it as well by adding machine learning models.

Climate Vulnerability & Migration Patterns. Grecequet et al. (2017), in *"Climate Vulnerability and Human Migration in Global Perspective"* [Grecequet et al., 2017], analyzed how a country's climate vulnerability affects migration flows. Because climate vulnerability is somewhat subjective, one of their goals was to turn the idea into a numerical score. What they found is that people tend to migrate from countries with high climate vulnerability scores towards the lesser ones. This makes sense, and it suggests that migration can function as a risk reducing strategy. The authors also talked a lot about 'trapped' populations, which are highly vulnerable groups that just cannot move because of economic or social constraints. This study was informative, but it does not make predictions. It identifies correlations but does not attempt to forecast migration.

Environmental Change & Migration Framework. Black et al. (2011), in their influential article *"The effect of environmental change on human migration"* [Black et al., 2011], introduced a new framework that describes the relationship between environment and migration through five main points: economic, social, political, demographic, and environmental. The authors say that climate change usually is not the main force for migration. It amplifies other drivers of existing mobility. Environmental stress can either directly or indirectly affect livelihoods and conflict. The authors stress that not all populations are able to relocate, similar to the previous paper. Our work compliments this framework by adopting a data driven approach: using historical climate, disaster, and displacement data to predict migration outcomes.

3 Methods (1-2 pages)

3.1 Dataset(s) & Preprocessing

Due to the fact that climate induced displacement depends on a lot of factors, we combined three major global datasets. Each of them contributed a unique perspective and helped our models understand.

- **Global Refugee & Migration Dataset (UNHCR).** We chose this dataset because we needed counts of displaced populations. It contains counts of refugees, asylum seekers, internally displaced persons, and other groups of concerns. Put simply, it tracks who leaves, where they go, and when. This dataset enables us to see migration flows, and we aggregated all displacement categories into a single continuous variable, *Total_Displacement* for each country-year.

- **Disaster Displacement Dataset (IDMC/GIDD).** The Disaster Displacement Dataset from IDMC and GIDD records internal displacements that were the result of specific natural disasters. These could be storms, floods, earthquakes, or droughts. We wanted to include this because it provides direct human impact for climate hazards. From over 13,000 documented disaster events, we engineered several features: total disaster-related displacements, the number of disaster events, the maximum displacement from a single event, and annual counts of each hazard type (e.g., floods, storms, wildfires).

- **Global Climate Dataset (NASA/CRU).** This dataset is gridded, which is usual for large climate datasets, and includes monthly climate data (temperature, precipitation, etc.) across a global grid. Since we needed to standardize all three datasets to use ISO3 codes and have a year column, we had to process this dataset heavily. The first step was to map each grid cell to its nearest country, using GeoPandas. Then, we aggregated monthly measurements into annual per-country summaries. From these new features, we created more such as temperature and precipitation deviations from historical long term means to capture any anomalies.

3.2 EDA & Feature Engineering

Before using the data for our models, we explored the data to understand the relationships between features and their distribution:

- **Trend Analysis:** We explored time-series plots that showed global displacement and found a positive trend in total displacements over the past 20 years, which aligns with more frequent disasters and climate stresses. We observed some years with spikes that correspond to major disaster event outliers, causing mass displacement.
- **Distribution of Displacement:** The distribution of annual displacement across country-years was very skewed. Several country-years, such as 2010, had high displacement numbers due to major disasters, like the 2010 Haiti earthquake, while most others were much lower. Due to this skewed distribution, we must normalize or transform the data to prevent the model from being strongly influenced by rare major events.
- **Climate Feature Patterns:** We performed Principal Component Analysis, PCA, on climate features to catch the strong patterns. The first few principal components explained more than **90%** of the variance in climate data. For example, **"Climate_PC1"** practically captures an overall "warm/wet" and "cool/dry" gradient, which distinguishes between hot/wet years and cold/dry years, while **"Climate_PC2"** captures variability patterns, distinguishing years with major fluctuations in temperature or precipitation.

Based on EDA knowledge, we engineered several features:

- **Climate Principal Components:** We retained the top 5 PCA components from the climate data to use as features. This reduced multicollinearity between the raw climate features and provided us with a visualization of climate anomalies each year.
- **Hazard Type Counts:** For each major climate disaster, such as storms, floods, droughts, etc., we included the annual count of these occurrences per country. This allows the model to learn whether certain hazards, such as floods, have a stronger impact on displacement compared to others. A country that experiences 5 floods may have a different displacement result than one with one major earthquake, for example. Therefore, the model will weigh the features appropriately.
- **Population Normalization:** We merged each country's population to normalize displacement relative to population size. For some analyses, we looked at displacement as a fraction of the population; however, for modeling, we primarily used absolute numbers with population used as a feature. Including population as a feature is important to help the model understand exposure, as displacement is also dependent on how populous a country is.
- **Log Transformation of Target:** Due to the skewed distribution of the **"Total_Displacement"** feature, we applied a log transform to the target variable. All the models are trained in log-space, which stabilizes variance and decreases the influence of outliers. A difference of $+0.7$ in log-displacement reflects about a 2x difference in actual displacement, which our models can handle.

3.3 Models

Predictive Models: Since we are predicting log-displacement, we planned to use regression to produce our predictions. As such, we implemented three regression-based machine learning models:

3.3.1 Model 1: [Cart/Tuned Cart]

- **Cart(Decision Tree):** We used a single Classification and Regression Tree as a base model. Due to their process of recursively splitting the data, decision trees are easy to interpret and show nonlinear relationships. Our first CART model, which used default parameters, was untuned to gauge the baseline performance.

  Additionally, we did a tuned CART model with "Grid Search" for the best depth/split parameters; however, the tuned tree did not generalize better than the untuned model. We inferred that this result is due to overfitting of the training data and the limitations of single-tree regression.

3.3.2 Model 2: [Random Forest]

- **Random Forest:** This ensemble method creates many decision trees, each on a bootstrapped sample of the data, that consider a random subset of features at each split and average their predictions. Due to randomness and averaging, this improves stability and reduces overfitting compared to a single tree. We trained a Random Forest model with default parameters as our strongest model. We expected that the forest model would capture the complex feature relationships and perform much better than our CART Model.

3.3.3 Model 3: [XGBoost (Gradient Boost)]

- **XGBoost (Gradient Boosting):** We also implemented an XGBoost Regression model, which sequentially builds trees, with each tree correcting errors of the previous one. From optimizing residual errors, Gradient boosting is typically highly accurate if tuned correctly to avoid overfitting. We tried some default boosting parameters with hyperparameter tuning, such as learning rate and max depth, based on validation performance. Like Random Forest, XGBoost can capture nonlinear interactions and model more complex data.

3.4 Training and Evaluation Methodology

- **Training Procedure:** A critical aspect of our approach was using a time-based train/test split. Instead of random shuffling of data, we trained the models on data from the years 2000-2015 and tested them on the years 2016-2019, the most recent data available, to simulate true predictions. This reflects the chronological order, ensuring the model predicts "future" year displacement. We did not perform standard cross-validation across all years, since this could reveal data from future years. Even our Grid Search for the CART models was carefully tuned to prevent leaking data after 2015. Model performance was evaluated on the test set using the $\mathbf{R}^2$ and **RMSE** scores, mostly computed in log-space. To interpret the results, we converted predictions back to original units to see actual displacement numbers, but the error metrics were most reliable in log space due to the large values we worked with.

4 Results (0.5-1 page)

4.1 Model Evaluation

- **Overall Model Performance:** In terms of model accuracy, the ensemble models significantly outperformed the single decision tree.

  The Random Forest achieved an R^2 score of around 0.61 in logarithmic displacement and an RMSE score of ≈ 1.75, closely followed by XGBoost with an R^2 score of ≈ 0.61 and RMSE score of ≈ 1.79. Both models explained $\approx 60\%$ of the variance in log-transformed results, indicating a strong predictive model.

  Contrasting that, the R^2 score had an R^2 score of ≈ 0.42 in log-space. This reflects the strength of ensembling, as averaging many trees allows Random Forest to capture patterns that single trees could not.

  Interestingly, with raw displacement units, the untuned CART had an R^2 of ≈ 0.33, slightly higher than its log-space R^2, but this was misleading due to its tendency to predict median values. The tuned CART did not improve from the base CART model. In fact, it performed worse with an R^2 of ≈ 0.23 in log-space, likely because of the tuning process, which resulted in overfitting.

- **Importance of Log-Space Evaluation:** During our process of evaluating predictions, we found that evaluating in the original scale of total displacement misrepresented the true numbers. For example, in the Random Forest's RMSE score in raw units was about 900,000 people, compared to ~ 1.75 in log terms. Therefore, a single huge event could heavily influence the RMSE score. Thus, using log-space metrics provided a more accurate evaluation of model accuracy, to prevent major disasters from dominating error. Moreover, we used the log-scaled R^2 scores to compare models, as they were more significant: it showed that Random Forest and XGBoost captured over 50% of variance, compared to the low raw R^2 score. Thus, the analysis of our results is based on log-space performance.

- **Key Predictors:** We explored the Random Forest's learned feature importances helped us understand feature patterns. Among the top predictors were Population, the total population of a country, and Total Disaster Displacements, the number of people displaced by disasters during that year. This aligns with our predictions, as larger countries and those already experiencing disaster-induced displacement tend to have higher overall displacement.

  Climate-related features also ranked highly: for example, Climate_PC1, the climate anomaly component, was significant, indicating that years with unusual temperature/precipitation patterns correlate with migration spikes. Certain hazard-type counts, especially storms and floods, had high weights as well, aligning with the expectation that events like tropical cyclones or severe floods drive people from their homes. Alternatively, features like drought count had less importance. This extra knowledge gave us confidence that the model is considering meaningful signals like climate extremes, disasters, and population exposure, rather than other noise data in its predictions.

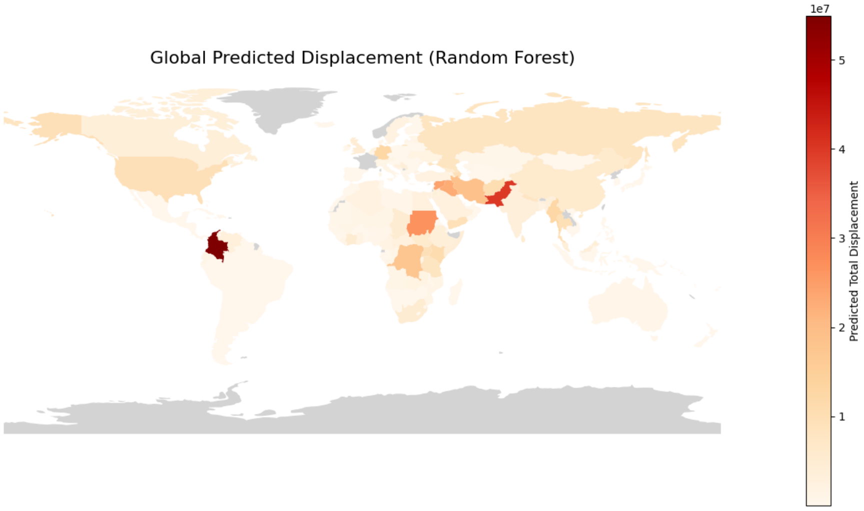

Figure 1: **Global Predicted Displacement Map**

Warmer colors, towards red, indicate higher predicted displacements for 2019. Countries in Asia and Africa—such as Pakistan and Ethiopia—exhibit the highest predicted values, reflecting both high population exposure and frequent climate-related disasters such as floods and droughts in the late 2010s. Overall, the spatial clustering aligns with known climate-vulnerable regions, suggesting the model captures meaningful geographic trends.

- **Visualization of Results:** In addition to the map above (Figure: 1), we also generated other visualizations to interpret the model's behavior. For example, we plotted predicted vs. actual displacement for test data. The Random Forest's points clustered along the diagonal in log-space, meaning good calibration except at the very high end, where it under-predicted the most extreme actual outcomes. We also visualize displacement by continent, combining our country-level predictions to understand continental or regional total displacement. Asia and Africa showed the highest numbers of displacement, which is consistent with continents of large populations and higher climate vulnerability. Thus, these visualizations help communicate the scale of predicted displacement in different parts of the world.

4.2 Comparative Analysis

- **Metrics by Model:** In log-space, Random Forest achieved $R^2 \approx 0.63$ and RMSE ≈ 1.75, XGBoost $R^2 \approx 0.61$ (RMSE ≈ 1.79), and CART $R^2 \approx 0.42$ (RMSE ≈ 2.16). These indicate the ensemble models explaining $\sim$ 55–60% of variance in log-displacement, versus $\sim$ 42% by the single tree. Therefore, the Random Forest's predictions had an average error of $\sim \pm 1.7$ in population displacement. Thus, level of accuracy helps distinguish between a 50,000 displacement event versus a 500,000 event, for example. The decision tree often under-predicted the extreme events, due to averaging, highlighting why it achieved a seemingly decent raw R^2 score. Since it never guesses huge values, it avoided giant errors, but at the cost of missing the major events altogether.

5 Discussion (0.5-1 page)

5.1 Connections to Prior Work

Our findings are consistent with the studies on climate impacts and migration presented earlier in this paper.

Benevolenza and DeRigne (2019) found that the most populations affected are those with fewer resources and who are highly vulnerable due to climate-related disasters. While our resulting data relates the migration to a country, we still see a similar trend that the countries where our model predicts the most displacement are often low-income regions that face repeated hazards. This supports the notion that exposure, and limited capacity to adapt, makes certain populations more likely to be displaced.
Grecequet et al. (2017) constructed a climate-vulnerability index and found that individuals migrate from the more vulnerable to the less vulnerable countries. Our findings align well with these. Even though we do not explicitly use their index, it turns out that many of the same high-risk areas are also identified by our model. In addition, instead of only showing the correlations, our work goes further by making actual predictions about displacement based on climate and disaster trends.
As Black et al. (2011) describe, environmental factors cause migration, however are not the sole reason of migration. It's also shaped by social, economic, and political factors as well. Another consideration they note is "trapped populations"-people who are highly exposed to climate risks but unable to move. Our models reflect this idea, too. While climate and population features are helpful in allowing us to anticipate displacement, the model cannot explain why some of the most vulnerable countries report relatively low displacement. This might be because people simply lack the wherewithal to move, not because the risk is low.
Overall, these similarities demonstrate how our work is situated within the research already undertaken in this field.

5.2 Limitations

While our model produces useful data, it has several important limitations:

- **Country-level data is too broad.** Each data point represents an entire country in a single year. This significantly simplifies the origin of the migration. A disaster might affect one country's region much more than another, and for countries as big as Russia or Canada the resulting data is too broad.

- **Missing social and economic factors.** We only use climate data, hazard information, and population. We didn't include things like income, conflicts (although the migration data often can be directly mapped to a specified conflict), or government stability, which are known to influence migration. Because of this, the model may miss key reasons why displacement varies from place to place.

- **Data quality issues.** The displacement data is not perfect. Some countries may not report all of their displacement events. We also had to fill in some missing values, which also adds uncertainty.

- **Difficulty predicting extreme events.** Even though log scaling helped, the model still tends to underestimate very large disasters. These exceptionally rare, but very important events drive significant migration. However they're difficult to predict with the past data alone.

- **Changing climate conditions.** Our model assumes that the patterns seen in the years 2000–2015 will continue into the future. However, climate change keeps increasing the frequency and severity of extreme events, meaning that the model won't work as well when climate conditions worsen and the frequency of disasters increases.

Because of these limitations, the predictions should be used carefully. They are helpful for understanding general trends, but not for making precise forecasts about extreme or unusual events.

6 Conclusion & Future Work (0.5 page)

In this project we combined global climate, disaster, and displacement data to build a model that predicts the displacement each year in different countries. Tree-based models like Random Forest and XGBoost performed the best, explaining around 60% of the variation in displacement when using a realistic time-based split. Our models found that population size and climate patterns were the strongest predictors, and the maps of predicted displacement matched well with known high-risk regions.

Overall, our results show that standard machine-learning methods can capture real and meaningful patterns in climate-related displacement. It can help us better understand which countries are most at risk and why, although they cannot fully replace deeper social, political, and economic analysis.

Going forward there are multiple things that could be done to improve the model's accuracy:

- **Add more types of data.** Things like income, conflicts, government quality, and urbanization could help the model understand the full picture of why people migrate.
- **Use more detailed data.** Predicting displacement for smaller regions within each country could show much richer patterns.
- **Incorporate future climate scenarios.** By implementing the predictions of how climate change is going to change in the future, the model would be able to predict the displacement more realistically.
- **Check fairness and accuracy across regions.** Some areas may be harder for the model to predict than others. Understanding where the model struggles can help improve reliability.

By implementing these steps the model's accuracy would increase making the predictions more realistic and useful for real-world planning.

Acknowledgments

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